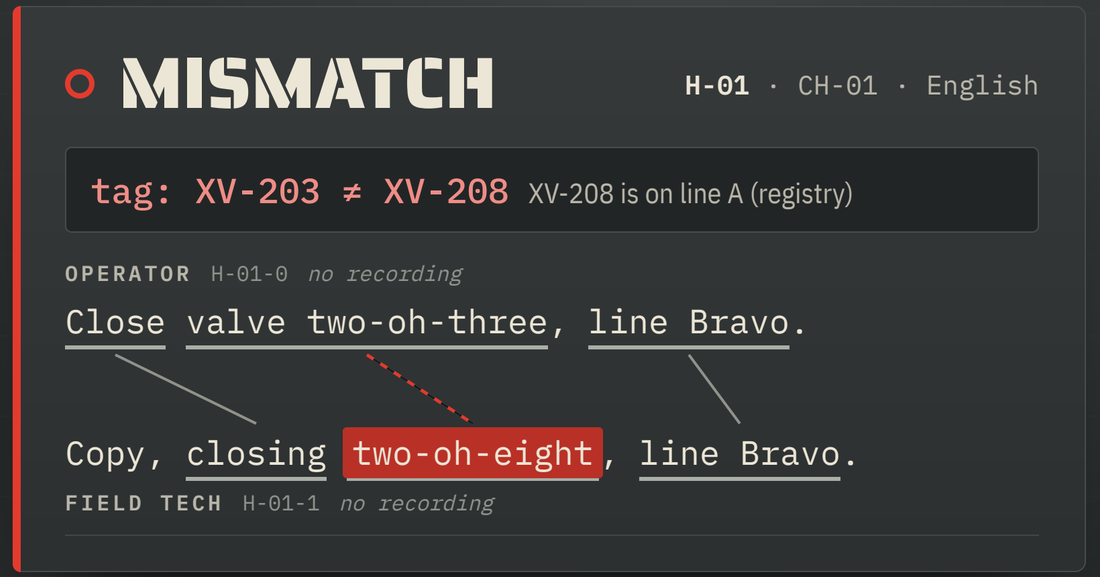
SAYAGAIN

NOBODY ON THE CHANNEL NOTICED. THE RADIO LOG DID.

Checks every radio read-back of a safety instruction, and sends the wrong ones to a human.




```
zsh · build/
$ uv run python scripts/eval.py --rescore
OK ladder-r1-reaudit.json :: Gemma-4-31B-IT (rows, calls, meets_rule)
OK llm-judge-r1-reaudit.json :: llm_judge (rows, calls, meets_rule)
```

CH-01 · HOW THE CHANNEL HEARS IT TODAY

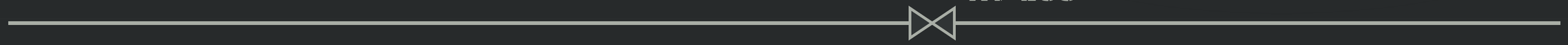
TWO LINES. NOBODY FLAGS THEM.

OPERATOR — Close valve two-oh-three, line Bravo.

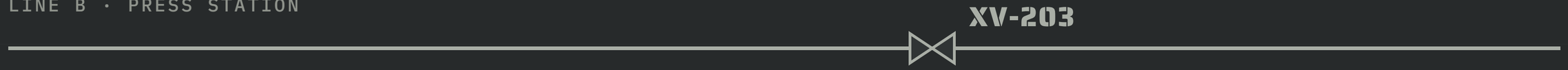
FIELD TECH — Copy, closing two-oh-eight, line Bravo.

no flag · no card · the crew walks to XV-208

LINE A · CLARIFICATION



LINE B · PRESS STATION



In the registry the demo runs on, XV-203 is on line B at the press station; XV-208 is on line A, in clarification.

THREE-WAY COMMUNICATION

READ-BACK IS THE CHECK. NOTHING CHECKS IT.

63 % - 50 %

of read-back errors caught by controllers

Helmke 2021 · aviation

1 killed, 6 injured

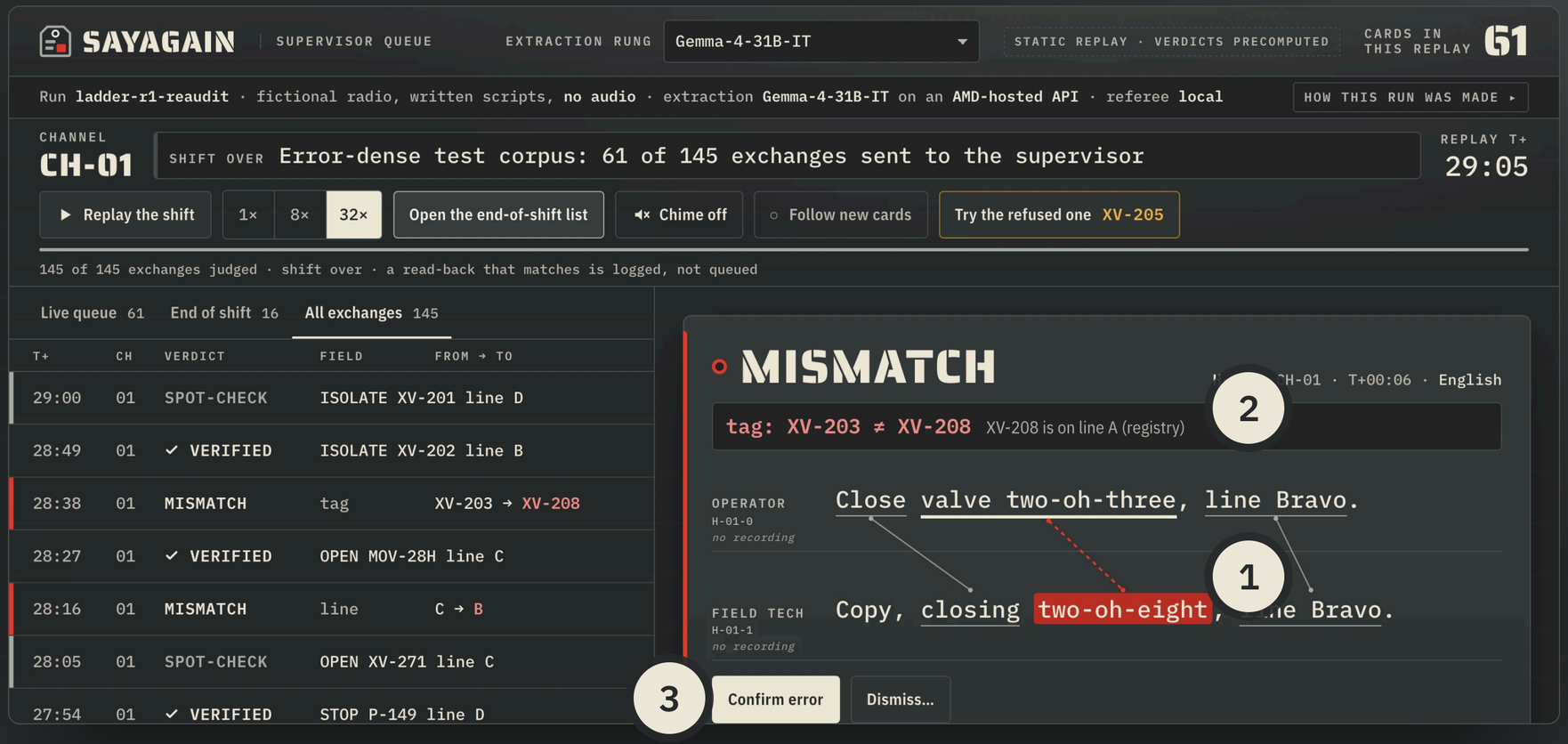
isolation valve left closed

Goodyear 2008 · CSB

No rate

published for plant radio

THE SUPERVISOR GETS THE ONE WRONG WORD.



1. the read-back word that differs
2. the diff, checked against the registry
3. a human confirms or dismisses

ONE MODEL JOB · EVERYTHING ELSE IS CODE

THE MODEL POINTS. CODE DECIDES.



VERIFIED · MISMATCH · NO-READBACK · ABSTAIN:
written by code, never by the model

JUDGING IS NOT THE MODEL'S JOB.

LLM-as-judge · DeepSeek-V4.1-Flash, no registry

~~VERIFIED~~

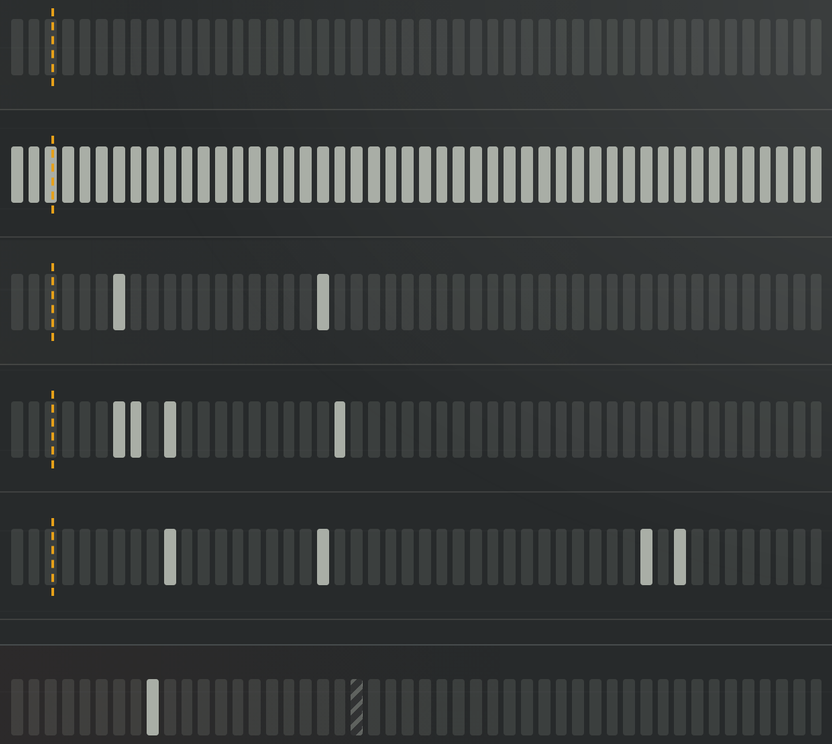
passed the XV-205 order, and 2 wrong read-backs



MODEL LADDER · 89 HELD-OUT EXCHANGES · TEXT·SCRIPTED

CLIMB ONLY AS HIGH AS NEEDED.

	false-VERIFIED	false alarms on 48 correct read-backs · dashed: the pre-registered budget		
regex rung 0 · no model	0		0/48	meets the rule
MiniCPM5-2B smallest model	0		48/48	
Gemma-4-31B-IT thinking off	0		2/48	meets the rule
DeepSeek-V4-Flash thinking off	0		4/48	
Qwen3.8-27B thinking on	0		4/48	
LLM-as-judge baseline · no registry	2		1/48	passed XV-205



On written words, rung 0 (a frozen regex) already meets the rule. The audio rows decide whether a model earns its place.

RECEIPTS · WHERE EACH STAGE RAN

EVERY STAGE SAYS WHERE IT RAN.

speech-to-text	span extraction	referee	queue
not measured yet Oct 16	AMD Token Factory · developer.amd.com.cn a receipt per call	local · plain Python 0.77 ms on H-01	local · FastAPI reviews in SQLite

11
instruct models probed

8
read the read-back as CLOSE · XV-208 · line B

0
accepted JSON-schema output

0
transcription models

WHO PAYS

PRICED PER CHANNEL. BOUGHT BY HSE.

1,318	mills, Indonesia 2022
× 30 %	addressable*
× 6	channels per mill*
× USD 2,400	per channel-year*
≈ USD 5.7 M	a year, mills only

* assumption

BUYER

HSE / K3 manager (SMK3 audit)

PUBLIC DOCUMENTATION, READ OCT 2026

LOGGERS TRANSCRIBE. NONE WE CHECKED COMPARES.

	TRANSCRIBES RADIO	COMPARES A READ-BACK	PLANT TAG REGISTRY
NICE Inform AI	**yes**	not found	no
Eventide NexLog + Speech Factor AI	**yes**	not found	not found
Motorola Assist	**yes**	not found	not found
IntelliPERMIT	no	radioed codes, not speech	**yes**
HAAWAII (air-traffic research)	**yes**	**yes, aviation**	no
SayAgain	planned	**yes**	**yes**

HONEST BY DESIGN.

> “No recorded speech has been measured.”

> “The plant, its 120 tags and every exchange are fictional”

> “...what a language model adds is unproven until the audio rows exist.”

MEASURED OCT 16

audio row	failure case	kill test	auditor

PUT IT ON ONE MILL'S RADIO.

Q1 2027	paid pilot, one palm-oil mill
Q3 2027	DCS check: did the valve close?
Q4 2027	Malaysian palm belt



github.com/edycutjong/sayagain

public at submission

EDY CU · JAKARTA